A Major Project Report

(Phase-I)

On

AI-BASED SIGN LANGUGAGE TRANSLATOR

Submitted in partial fulfillment of the requirement for the award of the degree of

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CANDIDATE'S DECLARATION

We hereby certify that the work presented in the **Phase-I report** of the project entitled "***AI Based sign language translator***", in partial fulfillment of the requirements for the degree of **Bachelor of Technology** in the **School of Computing Science and Engineering, IILM University**, Greater Noida, is an original work carried out by us under the supervision of ***Ms. Sonam Sirohi*** , School of Computing Science and Engineering, IILM University, Greater Noida.

The work presented in this Phase-I report has not been submitted by us for the award of any other degree or qualification at this or any other institution.

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This is to certify that the above statement made by the candidates is correct to the best of my knowledge.

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ABSTRACT

Sign language is the main way of communication for many Deaf and Hard-of-Hearing people around the globe, but the lack of a universal sign language and limited availability of human interpreters creates a challenge in day-to-day conversations, particularly in medical institutions, schools, and government offices. Current automatic sign-language recognition (SLR) technology is typically narrow, being programmed to understand one specific language (mostly American or Irish Sign Language) using a limited set of lexical items. This project will develop a language-agnostic real-time recognition system that uses a modular approach allowing the easy introduction of new sign languages by replacing or adding a recognition module trained on the desired language. The system will leverage MediaPipe's hand, pose, and face landmark trackers and utilize a hybrid CNN-LSTM architecture to recognize gestures. The CNN part will extract features from individual frames, while the LSTM layer will learn the sequence recognition task. The output text can be synthesized into speech in real-time and viewed in a web or desktop application. The first phase of the project will outline the problem statement, review the recent approaches to SLR, including CNN-, LSTM-, and Transformer-based models, identify the language-agnostic design as the primary research question, and evaluate the technical, economic, and operational feasibility of the proposed solution. A prototype will be designed to achieve at least 85% recognition accuracy on a reduced set of lexical items with less than 300 ms of end-to-end latency and implement the model in a web/desktop app with confidence-based output and options to add other sign languages in the following iterations.

AI-BASED SIGN LANGUAGE TRANSLATOR

A Real-Time, Multi Sign-Language Recognition System using Computer Vision and Deep Learning

Major Project — Phase I Report

Problem Definition, Literature Survey, Objectives, Feasibility Study, Scope, Novelty, and Feasibility

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1. Problem Definition

Sign language is the primary mode of communication for millions of Deaf and Hard-of-Hearing (DHH) people around the world. However, most hearing people do not understand any sign language, which creates a persistent communication gap in everyday situations such as hospitals, schools, banks, airports, and government offices. Human interpreters exist, but they are expensive, not always available on demand, and cannot scale to the millions of individual conversations that happen daily.

Sign languages are also not a single universal language. American Sign Language (ASL), Indian Sign Language (ISL), British Sign Language (BSL), and dozens of other national and regional sign languages each have their own grammar, handshapes, and non-manual markers (facial expressions, head tilts, mouth patterns). This means a solution built only for one sign language does not help a Deaf person who uses a different one, and existing automatic translators are almost always narrow, single-language systems trained on small, isolated-word datasets.

The core problem this project addresses is: How can a computer vision and deep learning based system automatically recognize hand gestures, body posture, and facial expressions from live video and translate them into readable text or speech, in a way that can be extended across multiple sign languages rather than being locked to just one?

Sub-problems

- Real-time gesture recognition: capturing and classifying fast, continuous hand and body motion from a webcam feed with minimal delay.
- Vocabulary and grammar gap: sign languages are not word-for-word translations of spoken languages; word order and grammar differ.
- Multi-language generalization: building a system architecture that is not hard-coded to one sign language and can be retrained or extended for others.
- Data scarcity: unlike spoken language text, large, labelled, diverse sign-language video datasets are limited and uneven across languages.
- Variability in signing style: different signers, hand sizes, skin tones, lighting, and camera angles all affect gesture appearance.

Who is affected

The direct beneficiaries are Deaf and Hard-of-Hearing individuals who need to communicate with hearing people who do not know sign language, in settings like classrooms, hospitals, workplaces, and public services. Indirect beneficiaries include interpreters (who can use the system to reduce routine workload), families of Deaf individuals, and institutions that are legally required to provide accessible communication.

2. Literature Survey

Recent research on sign language recognition (SLR) has moved from classical machine learning (SVMs, HMMs on hand-crafted features) toward deep learning, particularly Convolutional Neural Networks (CNNs) for spatial feature extraction and Long Short-Term Memory (LSTM) networks or Gated Recurrent Units (GRUs) for modelling the temporal sequence of a gesture. Several studies also combine CNNs with 3D convolutions, ResNet backbones, attention mechanisms, or MediaPipe-based landmark extraction instead of raw pixels, to reduce the effect of background and lighting variation.

Author(s) / Year	Approach	Dataset / Scope	Key Finding	Limitation
Sinha et al., 2023	CNN + LSTM hybrid	Real-time sign prediction	Improved accuracy over single-model baselines	Substantial computational overhead
Rehman et al., 2022	3D-CNN + LSTM	Dynamic hand gestures	Combined spatial-temporal features boost accuracy	High training cost, large model size
Kothadiya et al. (DeepSign), 2020/23	LSTM + GRU hybrid	IISL2020 (11 Indian Sign Language signs)	97% accuracy on ISL subset	Small vocabulary; not generalized to full ISL
Venugopalan & Reghunadhan	Deep Convolutional LSTM	Emergency ISL gestures (COVID-19 context)	Up to ~99% accuracy on benchmark sets	Domain-specific (medical emergency signs only)
Meshram et al.	Survey / Review	Indian Sign Language, multiple architectures	3D-CNNs excel at spatial features; attention-LSTMs excel at continuous recognition	Highlights need for lightweight, region-specific models
Zhang & Jiang, 2024	Survey of DL methods	Multiple sign languages	Consolidates CNN, RNN, LSTM, Transformer approaches	Notes persistent dataset scarcity across sign languages
Chiang Mai Univ. study, 2024	ResNet + LSTM (CRNN)	Video-based SLR	ResNet backbone mitigates gradient issues in deep CNNs	Traditional CNN-LSTM still prone to vanishing/exploding gradients

Key observations from the literature

- Hybrid CNN-LSTM (and 3D-CNN + LSTM) architectures consistently outperform single-model baselines such as plain CNN or plain RNN, at the cost of higher computation.
- Landmark-based pipelines (e.g., MediaPipe Hands/Pose) are increasingly preferred over raw-pixel CNNs because they are lighter, faster, and less sensitive to background clutter and skin tone.
- Most published systems target a single sign language (commonly ASL or ISL) and a limited vocabulary (10–100 signs); very few attempt a multi-sign-language or open-vocabulary system.
- Dataset scarcity and lack of standardization across sign languages is repeatedly flagged as the biggest open challenge, more than model architecture itself.
- Attention mechanisms and Transformer-based approaches (e.g., SignFormer, SignBERT) are an emerging direction for continuous sign language recognition, beyond isolated-word classification.

Research gap

Across the surveyed work, there is no widely adopted, openly available system designed from the outset to be language-agnostic — one that treats "which sign language" as a configurable/selectable input rather than a fixed model. This project's contribution is to design the pipeline (data collection, landmark extraction, CNN/LSTM classification) in a modular way, so that phase-wise extension to more than one sign language is a matter of adding a new trained model/dataset rather than rebuilding the system.

3. Objectives

The following objectives guide Phase I and set up the scope for later phases:

1. To study the structure of sign languages (handshape, movement, location, orientation, and non-manual features) and survey existing deep learning approaches for automatic sign language recognition.
2. To design a modular system architecture that separates video capture, landmark/feature extraction, gesture classification, and text/speech output, so that new sign languages can be added without redesigning the whole pipeline.
3. To collect or curate a labelled gesture dataset (starting with one sign language, e.g., ISL or ASL, for a limited vocabulary) using webcam video and MediaPipe-based landmark extraction.
4. To build and train a CNN model for spatial feature extraction from individual frames/landmarks, and an LSTM model to capture the temporal sequence of a gesture.
5. To develop a real-time inference pipeline using OpenCV that captures live webcam video and predicts the corresponding sign/word with acceptable latency (target: under ~300 ms per prediction).
6. To convert recognized signs into readable text and, optionally, synthesized speech, through a simple user interface.
7. To evaluate the system using standard metrics (accuracy, precision, recall, F1-score, confusion matrix) on a held-out test set and document performance limitations honestly.
8. To establish a roadmap for extending the system to additional sign languages and to continuous (sentence-level) recognition in later phases.

4. Scope, Novelty, and Feasibility Study

Scope of the Project

This section defines the functional and non-functional boundaries of the system so that expectations are explicit for evaluators and for the team across phases.

Functional scope (what the system will do)

- Capture live video from a standard RGB webcam and detect the signer's hands, upper body, and face in real time using OpenCV and MediaPipe.
- Recognize isolated signs (single words/gestures) from a defined vocabulary for one target sign language in the initial build, using a CNN+LSTM pipeline.
- Translate a recognized sign into on-screen text, with optional text-to-speech output.
- Provide a simple interface (web or desktop) so a non-technical user can start/stop the camera and view translated output.
- Be architected so that a new sign language can be added by plugging in a new trained model and dataset, without redesigning the capture, extraction, or UI layers.

Non-functional scope (how well it should do it)

- Target real-time performance: prediction latency under ~300 ms per gesture on a mid-range GPU/CPU.
- Target recognition accuracy of 85%+ on the held-out test set for the chosen vocabulary, benchmarked against the literature in Section 2.
- Usable with normal indoor lighting and a plain or moderately cluttered background, without requiring studio conditions.

Explicit boundaries (what the system will not do, in this phase)

- It will not attempt full, grammatically correct continuous-sentence translation across sign languages in Phase I–III (this is future work, see Section 6.3).
- It will not claim simultaneous support for every world sign language at launch; multi-language support is delivered incrementally, one trained model at a time, on top of the same architecture.

- It will not perform two-way translation (spoken/text language into animated sign language output) in the current build.

Novelty / Innovation

Most published sign-language-recognition systems (see Section 2) are narrow: a single fixed model trained end-to-end for one sign language and a small, fixed vocabulary, with the architecture itself tied to that language. This project's novelty is less about inventing a new neural network layer and more about how the system is engineered to be extensible and honest about its limits:

- Language-agnostic pipeline design: video capture, MediaPipe landmark extraction, and the UI/output layer are decoupled from the recognition model, so "which sign language" becomes a swappable model/config rather than a hard-coded assumption — directly addressing the research gap identified in Section 2.2.
- Landmark-first, not pixel-first, recognition: using MediaPipe hand/pose/face keypoints as CNN/LSTM input (rather than raw video frames) reduces sensitivity to skin tone, background, and lighting — a fairness and robustness improvement over many raw-pixel CNN baselines in the literature.
- Hybrid CNN+LSTM with an explicit real-time budget: unlike research systems optimized purely for offline accuracy, this project treats sub-300ms latency as a hard design constraint from the start, trading off model complexity for usability.
- Confidence-aware output: rather than always forcing a prediction, the system is designed to expose a confidence score so a "not sure" result is visible to the user instead of silently mistranslating — relevant given the accessibility stakes described in Section 5.3.
- Extensibility as a first-class design goal: the modular four-layer architecture (Section 6.4) is itself the contribution — it turns "add a new sign language" from a rebuild into a configuration change, which is the practical innovation this project targets.

Feasibility Study and Analysis

A feasibility study was carried out across five dimensions — technical, economic, operational, schedule, and data feasibility — to confirm the project is achievable within the constraints of a final-year academic project.

Feasibility Dimension	Assessment	Risk Level
Technical	CNN/LSTM-based gesture recognition is well-proven in literature; OpenCV, MediaPipe and TensorFlow are mature, free, well-documented tools.	Low–Medium
Economic	Entire software stack is open-source; main cost is optional GPU compute and a webcam, well within a student project budget.	Low
Operational	System must run in real time on modest hardware; usability for a non-technical user needs UI/UX care.	Medium
Schedule	Multi-sign-language scope is ambitious for one academic year; needs phased scope reduction (start with one language, expand).	Medium–High
Data availability	Labelled datasets exist for ASL and a few national sign languages (ISL, BSL) but not uniformly for all sign languages worldwide.	High
Legal / Ethical	Use of publicly licensed datasets and consented video capture; no major regulatory barriers for an academic prototype.	Low

Technical feasibility

All core technologies required — Python, OpenCV, MediaPipe, TensorFlow/Keras — are free, open-source, well-documented, and widely used in published SLR research (see Section 2), which lowers technical risk. A

single mid-range GPU (or free-tier cloud GPU such as Google Colab) is sufficient to train CNN/LSTM models on a limited-vocabulary dataset.

Economic feasibility

Because the software stack is entirely open-source, the main real cost is an optional webcam and, if local hardware is insufficient, a small amount of cloud GPU credit. This keeps the project comfortably within a typical student budget (see Bill of Materials, Section 7).

Operational feasibility

The system is intended to run on a standard laptop with a webcam, without specialized hardware like depth sensors or data gloves, which makes it realistic to demonstrate and operate. A simple web or desktop interface (Flask/Streamlit) keeps it usable for a non-technical evaluator.

Schedule feasibility

Supporting "every sign language" simultaneously is not realistic within one academic year, since labelled data does not exist uniformly for all sign languages. The recommended approach is to build the full pipeline for one sign language and a moderate vocabulary in Phase I/II, validate it, and treat additional languages as a scalable extension (new dataset + retrained model plugged into the same architecture) in later phases.

Data feasibility

Public datasets such as WLASL (Word-Level ASL), INCLUDE (Indian Sign Language), and ISL-CSLRT provide a starting point for training and benchmarking, though vocabulary size and video quality vary and may require supplementary self-recorded data.

5. Identification of Multidisciplinary and Transdisciplinary Aspects

This project sits at the intersection of several disciplines, which is expected of a system meant for real-world accessibility rather than a purely algorithmic exercise.

Multidisciplinary aspects (knowledge drawn from multiple fields, applied largely separately)

- Computer Science / AI: computer vision (OpenCV), deep learning (CNN, LSTM), software engineering for the real-time pipeline and UI.
- Linguistics: understanding sign language grammar, phonology (handshape, movement, location, orientation), and non-manual markers is necessary to define correct labels and avoid mistranslation.
- Human-Computer Interaction (HCI) / Design: designing an interface that is usable by Deaf users and hearing users with no technical background.
- Electronics / Hardware: camera selection, frame rate and resolution trade-offs, and (in future iterations) possible sensor-based input such as depth cameras or wearable gesture sensors.
- Disability Studies / Social Science: understanding accessibility needs, etiquette, and lived experience of the Deaf community to avoid building a technically correct but practically unhelpful tool.

Transdisciplinary aspects (integrated collaboration to solve the shared problem)

Beyond simply borrowing techniques from each field, an effective sign-language translator requires these disciplines to inform one another directly: linguistic structure shapes how the dataset is labelled and how the LSTM sequence model is designed (since sign order and non-manual markers change meaning); Deaf-community feedback should shape which signs/vocabulary are prioritized and how errors are communicated to the user (silent failure is worse than a visible "not confident" signal); and HCI considerations (latency,

feedback, readability) directly constrain which deep learning architectures are practical for real-time use. This integrated, problem-centred collaboration — rather than each discipline working in isolation — is the transdisciplinary core of the project.

Societal and ethical considerations

- Bias and fairness: training data must include diverse skin tones, hand sizes, and signing speeds to avoid a system that works poorly for some users.
- Consent and privacy: video data of signers (including self-recorded data) must be collected and stored with informed consent.
- Accessibility-first design: the system should be evaluated with input from Deaf or Hard-of-Hearing users, not only measured by offline accuracy numbers.

6. Product Development Scope

Phase-wise scope

Phase I (current): Problem definition, literature survey, objectives, feasibility study, and system architecture design. Selection of one target sign language and vocabulary subset for the initial prototype.

Phase II: Dataset collection/curation, landmark extraction pipeline, and training of baseline CNN and LSTM models for isolated-word recognition on the chosen sign language.

Phase III: Real-time inference pipeline (webcam capture → prediction → text/speech output), UI development, and performance evaluation against target metrics.

Phase IV (stretch goal / future work): Extension to a second sign language to validate the modular architecture, and exploration of continuous (sentence-level) sign language recognition beyond isolated words.

In-scope for the prototype

- Isolated (single-word/sign) recognition from a limited, well-defined vocabulary (e.g., 50–100 common signs).
- Recognition using a standard RGB webcam only — no specialized depth or wearable sensors required for the core deliverable.
- Text output as the primary translation result, with optional text-to-speech as a secondary feature.
- A modular architecture where the trained model for a sign language can be swapped without changing the video-capture or UI layers.

Out of scope for the current phase

- Full continuous, grammatically correct sentence-level translation across sign languages (an open research problem).
- Simultaneous, truly universal support for "every" sign language at launch — this is treated as a long-term extension goal, not a Phase I/II deliverable.
- Two-way translation (spoken/text language back into animated sign language) is a possible future extension but not part of the current build.

High-level system architecture

The system is designed in four modular layers: (1) Input Layer — webcam video capture via OpenCV; (2) Feature Extraction Layer — MediaPipe-based hand/pose/face landmark extraction, reducing each frame to a compact numerical representation; (3) Recognition Layer — a CNN component for spatial feature learning per

frame and an LSTM component for learning the temporal pattern across a sequence of frames, together classifying the gesture into a sign label; (4) Output Layer — mapping the predicted label to text (and optionally speech) through a simple web/desktop interface. Each layer communicates through well-defined data formats so a layer (e.g., the recognition model for a new sign language) can be replaced independently.

7. Identification of Components and Bill of Materials (BOM)

The project is primarily software-driven; hardware requirements are minimal. The table below lists the software and hardware components required for Phase I–III, along with approximate costs (student/prototype scale, in INR).

#	Component	Type	Purpose in Project	Approx. Cost (INR)
1	Laptop / PC with GPU (GTX 1650 or better)	Compute (Hardware)	Model training and real-time inference	Existing / 45,000–70,000
2	External USB Webcam (1080p, 30fps)	Hardware	Capturing hand, face and body gesture video for input	1,200–2,500
3	Python 3.10+	Software / Runtime	Core programming language for the pipeline	Free
4	OpenCV	Software Library	Frame capture, pre-processing, image transforms	Free
5	MediaPipe (Hands/Pose/Holistic)	Software Library	Landmark and keypoint extraction from video frames	Free
6	TensorFlow / Keras	Software Framework	Building and training CNN and LSTM models	Free
7	NumPy, Pandas	Software Library	Numerical operations and dataset handling	Free
8	Public Sign Language Datasets (WLASL, INCLUDE, ISL-CSLRT, etc.)	Dataset	Training and benchmarking recognition models	Free (research use)
9	Cloud GPU credits (Google Colab Pro / Kaggle)	Compute Service	Additional training compute if local GPU is insufficient	0–1,000/month
10	Flask / Streamlit	Software Framework	Building the demo web interface for translation output	Free
11	Text-to-Speech engine (pyttsx3 / gTTS)	Software Library	Converting recognized text to speech output	Free
12	Storage (SSD, min. 512 GB) / Cloud storage	Hardware / Service	Storing datasets, model checkpoints, video samples	Existing / 500–1,500

Notes on the BOM

- Most software components are free and open-source; the only realistic recurring cost is optional cloud GPU compute if local hardware cannot train the models in reasonable time.
- Dataset cost is listed as free assuming the use of public research datasets (e.g., WLASL, INCLUDE, ISL-CSLRT) under their respective academic-use licenses; supplementary self-recorded data has no direct monetary cost beyond time.
- Hardware (laptop, webcam) is assumed to be largely available already to a student; costs shown are only for the incremental items that may need to be purchased.

8. Conclusion

Phase I has established a clear problem statement, grounded the project in current literature on CNN/LSTM-based sign language recognition, defined measurable objectives, and confirmed technical, economic, and operational feasibility for a phased, modular approach. The project deliberately narrows "every sign language" into a realistic, extensible roadmap: build a solid, modular pipeline for one sign language first, validate it against the stated objectives and metrics, and treat additional sign languages as incremental extensions of the same architecture in later phases.

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